Harpreet Singh

Research Assistant | PhD Computational Behavioural Neuroscience | Embedded AI / FPGA / Neurotechnology

Lethbridge, AB | +1 778-536-4298 | workwithharpreetsingh@gmail.com | linkedin.com/in/harpreet-singh-b42942213/ | github.com/baazSingh13

Professional Summary

Embedded AI and neurotechnology researcher with 13+ years across electronics design, firmware, FPGA acceleration, data engineering, cybersecurity AI, and biomedical research. Current Research Assistant and PhD student at the University of Lethbridge, building EEG/fNIRS acquisition, CGX/XDF dataset creation, live BCI classifier/game control, and temporal QiSNN FPGA pipelines for computational behavioural neuroscience. Strong record translating research ideas into working hardware/software systems, from schematics and PCB bring-up to Vitis HLS, Vivado, MicroBlaze/Zynq integration, Python analytics, and board-visible validation.

Technical Skills

Embedded & hardware	C, Embedded C, Python, Bash, ARM Cortex, STM32, AVR, PIC, 8051, ESP32, bare-metal firmware, FreeRTOS, Zephyr, Embedded Linux, PCB design, SMPS, board bring-up
FPGA & acceleration	Xilinx Vivado/Vitis HLS, MicroBlaze, Zynq Cortex-A9, AXI4-Lite, AXI HP, BRAM maps, UARTLite, Arty A7-35T/A7-100T, Arty Z7-20, fixed-point ap_fixed, HLS C/C++
AI/ML & data	PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, Bayesian/Markov networks, IDS models, NLP transformers, SNN/QSNN/QiSNN, biomedical signal analysis
Neurotech & acquisition	EEG/fNIRS acquisition, ADS1299, ADS8688, DAC8565, LSL, LabRecorder, CGX EEG, UDP markers, XDF sessions, real-time plotting, CSV/JSONL logging
Tools & protocols	Git, JTAG/SWD, oscilloscopes, logic analyzers, Wireshark, Zeek, UART, SPI, I2C, CAN, RS-485, Modbus, USB, Ethernet, MQTT, HTTP, WebSocket

Professional Experience

Research Assistant, Core-Hub Neuroengineering Solutions | *University of Lethbridge | Apr 2025 - Present*

- Build current PhD neuroengineering research stack for EEG/fNIRS acquisition, CGX/XDF dataset creation, live BCI classifier/game control, embedded device control, and FPGA temporal QiSNN acceleration.
- Developed an 8-channel EEG + 8-channel fNIRS Raspberry Pi platform using ADS1299, ADS8688, DAC8565, SPI/GPIO, TDM MUX sequencing, live plotting, and CSV logging for offline analysis.
- Built CGX/Sheeg workflows that launch CGX Acquisition and LabRecorder, record labelled visual-stimulus XDF sessions, preprocess EEG into CSV datasets, and export model artifacts for live classification.
- Developed closed-loop BCI/FPGA and rodent-decision pipelines using ADS1299 frames, MicroBlaze, BRAM windows, temporal QiSNN/SNN accelerators, UDP/UART intent packets, and five-state game control.

Research Assistant | *University of Regina | Jan 2021 - Apr 2024*

- Researched uncertain-reasoning IDS methods for DoS/DDoS detection using Bayesian Networks, Markov Networks, feature engineering, Zeek/Wireshark traffic analysis, and Python ML workflows.
- Published SECRYPT 2024 paper on an uncertain-reasoning intrusion detection system and presented work to an international cybersecurity research audience.
- Built additional ML research prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

Independent Embedded Systems Developer | *Self-Employed | Jun 2024 - Mar 2025*

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

Senior Software Engineer | *Planetcast Media Services Ltd. | Dec 2018 - Nov 2020*

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics/PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

Embedded Software Engineer / R&D Engineer | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd. | Feb 2016 - Dec 2018*

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and ATE tools; designed 15-20 W isolated SMPS sections with 78-81% measured efficiency.
- Performed board bring-up and firmware validation across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, GUI test automation, and industrial power-monitoring interfaces.

Research & Engineering Projects

QLIF_QSNN_FPGA | *QLIF + PC-DDM-SNN FPGA research for MNIST/N-MNIST*

- Created a self-contained software-to-hardware flow for PC-DDM-SNN and Quantum LIF models using 14x14 MNIST and 12-frame N-MNIST inputs packed as ap_fixed<16,6> BRAM words.
- Automated dataset-matched training, HLS weight export, software reference checks, Vitis HLS simulation/synthesis, Vivado block-design generation, XSA/bitstream production, and 1000-sample UART tests.
- Diagnosed UART accuracy logs, separating metadata/reporting defects from true classifier score vectors; documented rebuild boundaries to avoid overclaiming mismatched hardware-weight results.

QSNN_EarlyExit_FPGA_Research | *Artix-7 QSNN baseline and adaptive early-exit research*

- Built a reproducible Arty A7-35T QSNN MicroBlaze system with AXI4-Lite control, BRAM-backed input/output buffers, UART diagnostics, and class-by-class MNIST board tests.
- Recorded implementation evidence: 0.698-0.699 ms HLS latency at 100 MHz, 9.87% LUTs, 5.13% FFs, 41.0% BRAM tiles, 3.33% DSPs, 0 routing errors, and positive routed timing slack.
- Defined early-exit research methodology comparing ANN, SNN, fixed-step QSNN, and adaptive QSNN on accuracy, latency, power/energy, utilization, and robustness.

rodent_decision_qisnn_temporal and BCI_Game_Loop_FPGA | Temporal QiSNN decision pipelines

- Prepared a temporal QiSNN/SNN FPGA research pipeline for rodent decision prediction using DANDI-derived raster vectors, 12 time bins x 196 features, 2352-word Q10 input BRAM, and 3-class lick labels.
- Designed the FPGA/electronics branch of a closed-loop BCI stack where ADS1299 frames feed MicroBlaze input BRAM, an EEG-adapted active temporal QiSNN produces five intent scores, and Ethernet/Wi-Fi/UART/SPI transports drive a PC/phone game.
- Connected trained/quantized EEG intent models to a shared software/hardware contract with idle, left, right, up, and down classes, Q15 confidence scoring, golden-window replay, and safety gates before live commands.

arty_z7_50m_llm | *Zynq-based 50.35M-parameter LLM accelerator and voice-assistant proof path*

- Designed a staged Arty Z7-20 transformer accelerator contract with 16 layers, 512 width, 8 heads, 8,192 vocabulary, INT8/INT4 weight payloads, and DDR-resident model storage.
- Implemented an architecture where Cortex-A9 Linux handles tokenization/scheduling while programmable logic accelerates DDR-backed INT8 matrix-vector operations over AXI HP with AXI4-Lite control.
- Defined board validation gates including XSA/BOOT.BIN creation, 115200-baud UART proof tests, expected mismatches=0, 1,000 repeated launches, audio PWM, and microphone-to-response flow.

CGX_dataset_game, BCT_8EEG_8FNIRS, and Sheeg | Closed-loop BCI research stack

- Integrated ADS1299 EEG, ADS8688 fNIRS analog sampling, DAC8565 output control, watchdog PWM, TDM MUX sequencing, shared SPI0 locking, and Raspberry Pi GPIO concurrency into live acquisition scripts.
- Built CGX EEG dataset and game workflow: five-state visual-stimulus labels are recorded to XDF, preprocessed into labelled CSV, trained into classifier artifacts, and replayed live as INTENT UDP packets to a browser game.

Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRYPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.
- Published thesis: Singh H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. The University of Regina (Canada); 2024.

Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012